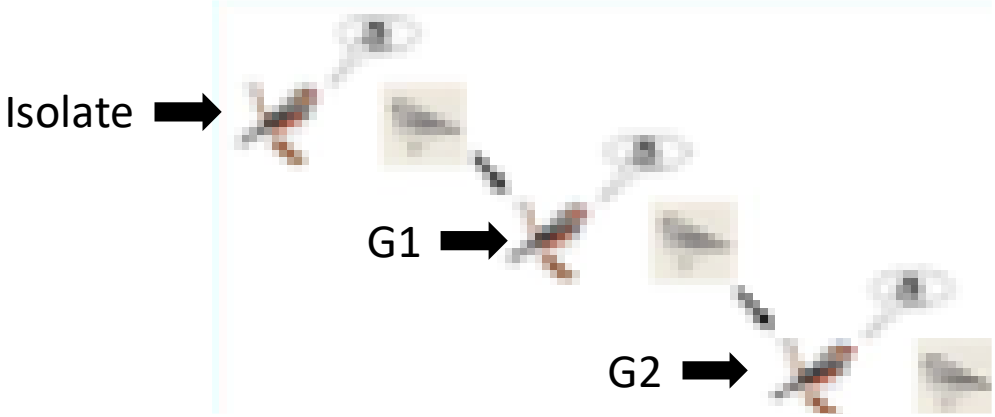
Origins and Evolution of Language
Week 7: (Grammar learning and then) The evolution of social cognition

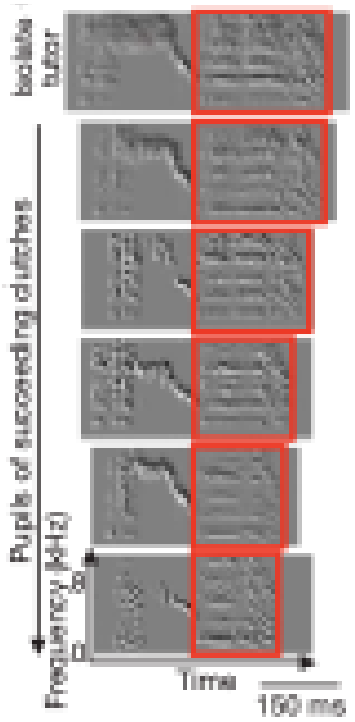
Kenny Smith

kenny.smith@ed.ac.uk

Follow-up from last week's tutorial: iterated learning in zebra finches







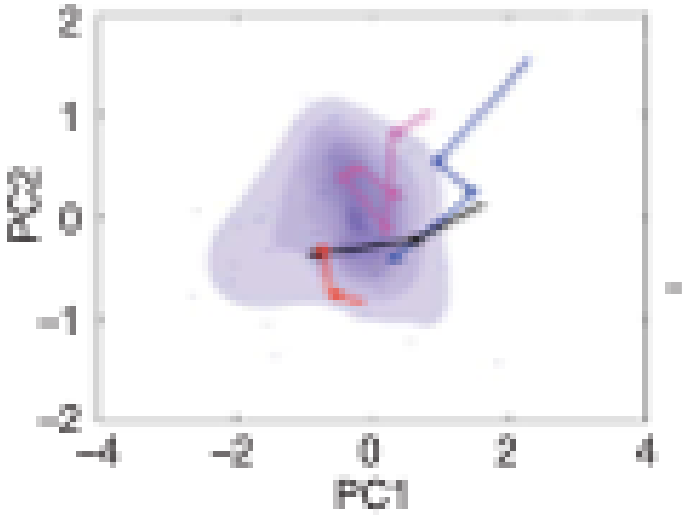
Fehér, O., Wang, H., Saar, S., Mitra, P. P., & Tchernichovski, O. (2009). De novo establishment of wild-type song culture in the zebra finch. *Nature, 459*, 564–568.

Follow-up from last week's tutorial: iterated learning in zebra finches



Isolate





Fehér, O., Wang, H., Saar, S., Mitra, P. P., & Tchernichovski, O. (2009). De novo establishment of wild-type song culture in the zebra finch. *Nature, 459*, 564–568.

An exam-related question from the pre-lecture quiz

“For the exam-style questions, should we always cite papers to support our ideas, or is it fine to use everyday examples?”

Plan for today

- Grammar learning from last week
  - And a reminder of where we are and where we are going
- Mind-reading and language
  - Mind-reading in word learning
  - Ostensive-inferential communication
  - Knowing what others know
  - The evolution of mind-reading

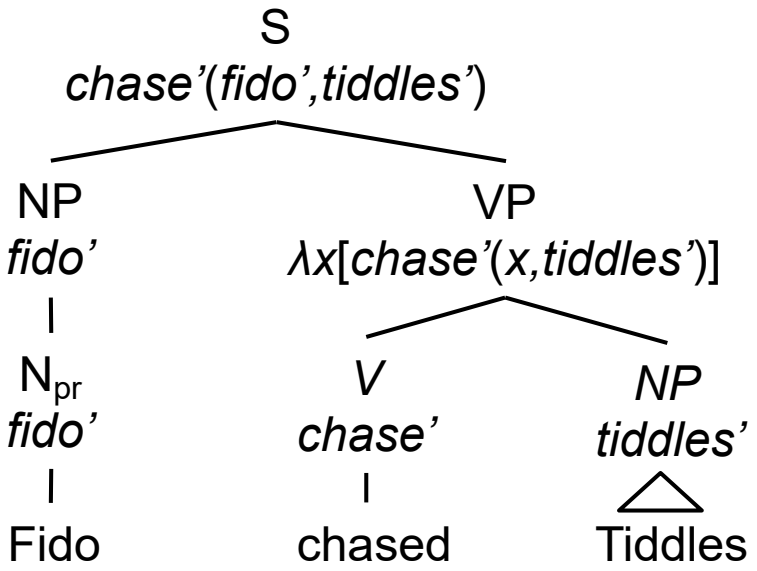
Grammar learning

See also this week’s tutorial

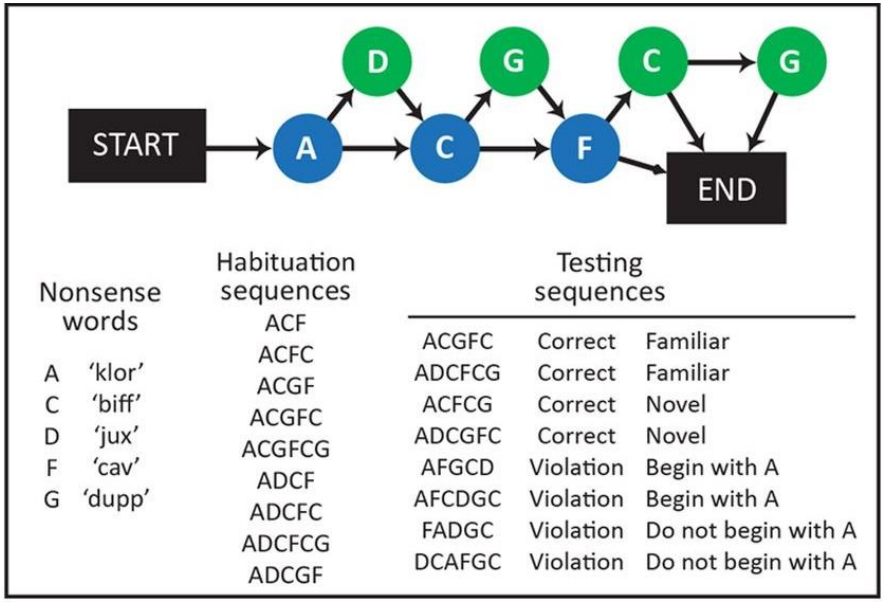
Reminder: Language's communicative power comes from its **structure**

Compositionality: the meaning of an expression is a function of the meaning of its parts and the way in which they are combined

S → NP VP	*VP'*(*NP'*)
NP → N_{pr}	N'_{pr}
N_{pr} → Fido	*fido'*
N_{pr} → Tiddles	*tiddles'*
VP → V NP	*V'*(*NP'*)
V → chased	*λx* [*λy* [(*chase'*(*x,y*)]]



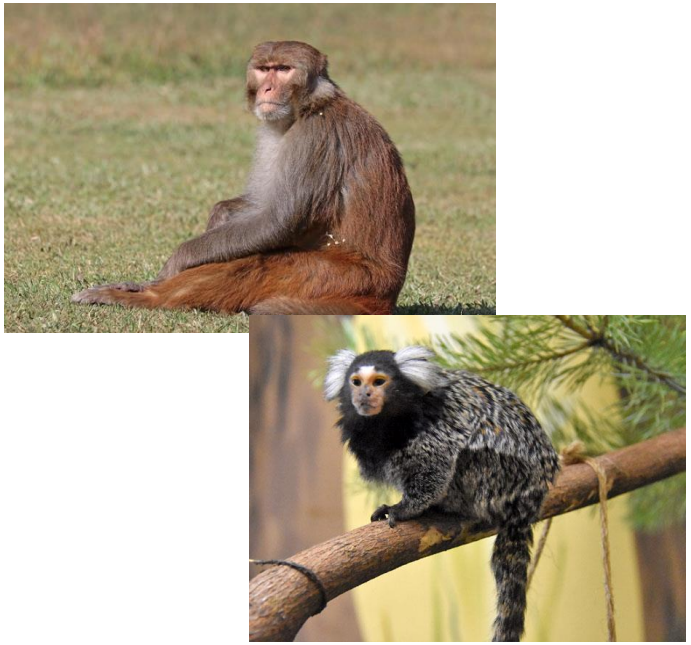
Artificial Grammar Learning in non-humans



Nonsense words	
A	'klor'
C	'biff'
D	'jux'
F	'cav'
G	'dupp'

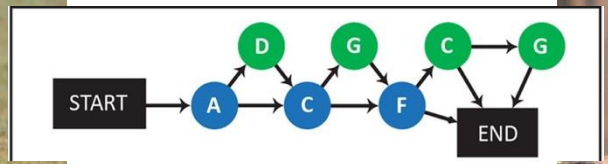
Habituation sequences
ACF
ACFC
ACGF
ACGFC
ACGFCG
ADCF
ADCFC
ADCFCG
ADCGF

Testing sequences		
ACGFC	Correct	Familiar
ADCFCG	Correct	Familiar
ACFCG	Correct	Novel
ADCGFC	Correct	Novel
AFGCD	Violation	Begin with A
AFCDGC	Violation	Begin with A
FADGC	Violation	Do not begin with A
DCAFGC	Violation	Do not begin with A

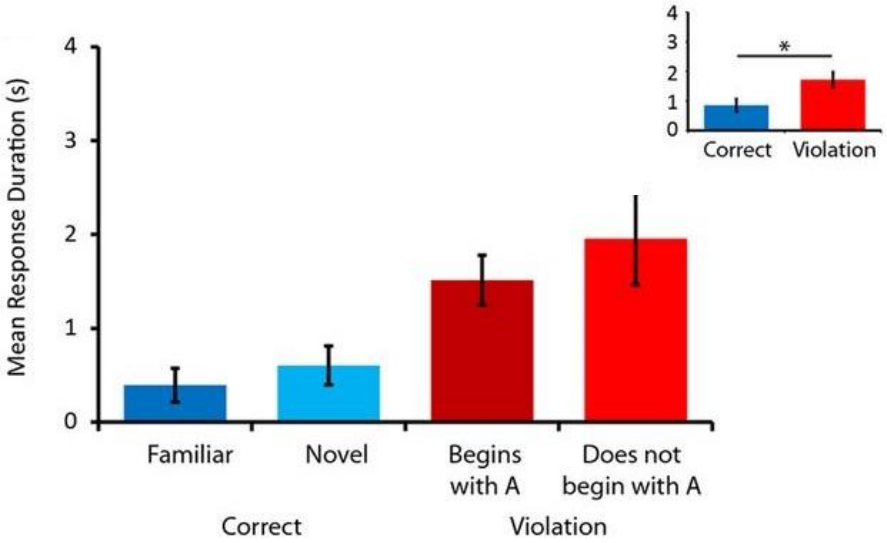


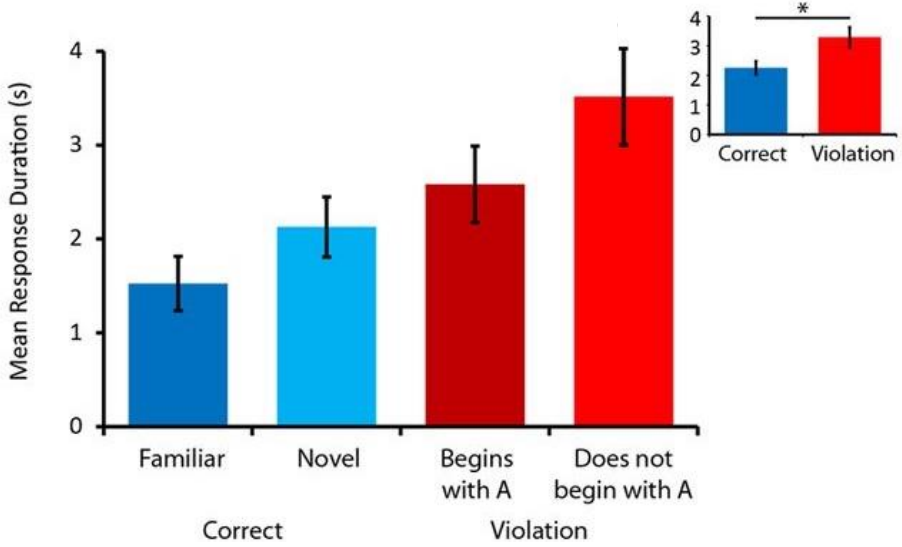
Wilson, B., Slater, H., Kikuchi, Y., Milne, A., Marslen-Wilson, W., Smith, K., & Petkov, C. (2013). Auditory artificial grammar learning in macaque and marmoset monkeys. *Journal of Neuroscience, 33,* 18825-18835.
For review see e.g. Petkov, C. I., & Ten Cate, C. (2020). Structured Sequence Learning: Animal Abilities, Cognitive Operations, and Language Evolution. *Topics in Cognitive Science, 12,* 828– 842.











Wilson, B., Slater, H., Kikuchi, Y., Milne, A., Marslen-Wilson, W., Smith, K., & Petkov, C. (2013). Auditory artificial grammar learning in macaque and marmoset monkeys. *Journal of Neuroscience, 33,* 18825-18835.

Non-adjacent dependency learning

Training

pel wadim rud… vot kicey jic… pel feenam rud…
vot wadim jic… vot puser jic… pel puser rud…

vot kicey jic… vot skiger jic… pel kicey rud …





Test

pel wadim rud

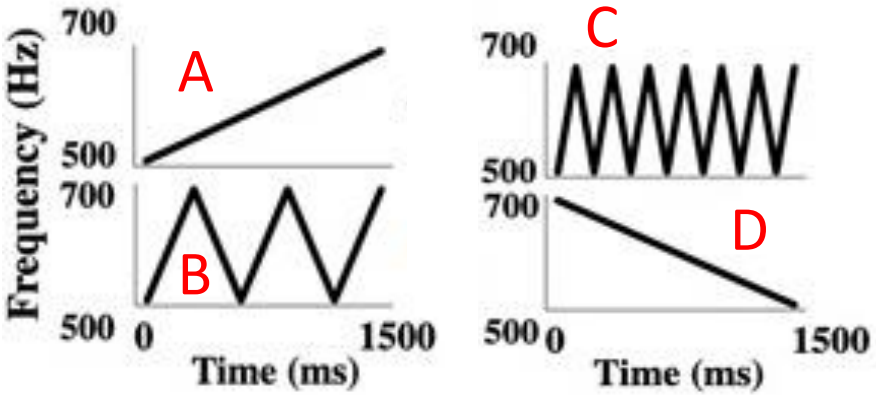
pel wadim jic?

vot wadim jic

vot wadim rud?

Gomez, R. (2002). Variability and detection of invariant structure. *Psychological Science, 13,* 431-436.

Non-adjacent dependency learning





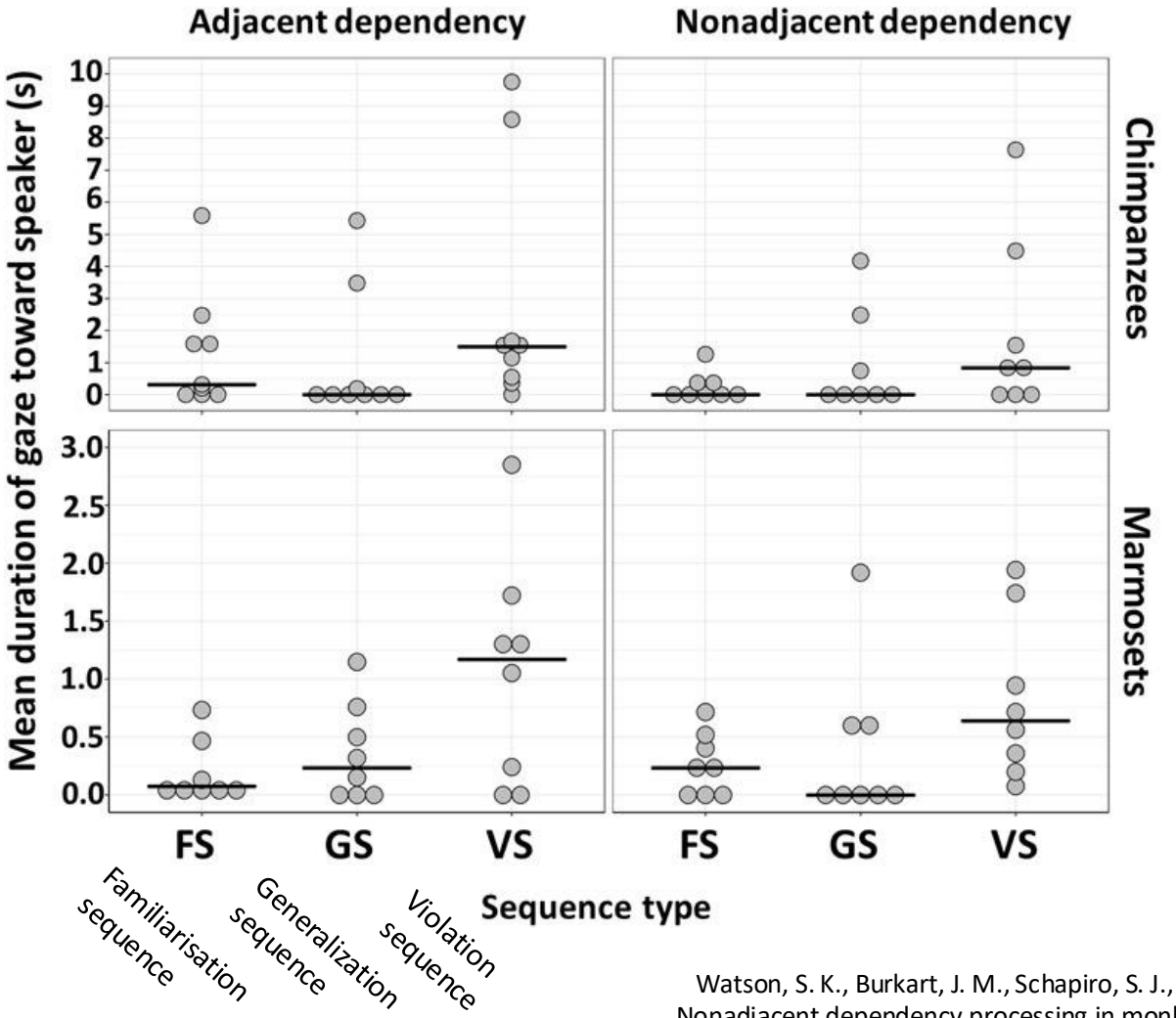
Watson, S. K., Burkart, J. M., Schapiro, S. J., Lambeth, S. P., Mueller, J. L., & Townsend, S. W. (2020). Nonadjacent dependency processing in monkeys, apes, and humans. *Science Advances, 6,* eabb0725.

Non-adjacent dependency learning

Sequence type	Example AD	Example Non-AD
Familiarization	$A_1 – B_7$	$A_1 – X2_4 – B_7$
	$C_5 – D_1$	$C_5 – X1_1 – D_1$
Generalization	$A_{13} – B_{11}$	$A_{13} – X1_{15} – B_{11}$
	$C_{10} – D_{16}$	$C_{10} – X2_{14} – D_{16}$
Violation	$A_{13} – D_{11}$	$A_{13} – X1_{15} – D_{11}$
	$C_{10} – B_{16}$	$C_{10} – X2_{14} – B_{16}$



Watson, S. K., Burkart, J. M., Schapiro, S. J., Lambeth, S. P., Mueller, J. L., & Townsend, S. W. (2020). Nonadjacent dependency processing in monkeys, apes, and humans. *Science Advances, 6,* eabb0725.







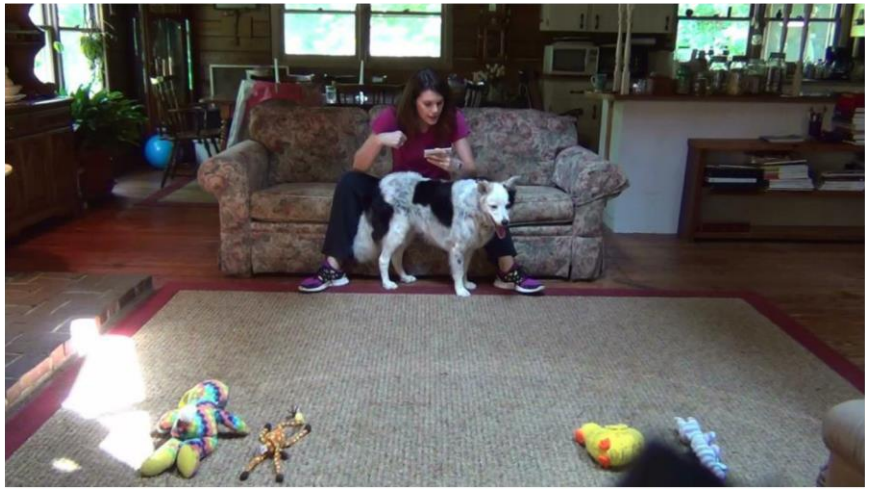
Watson, S. K., Burkart, J. M., Schapiro, S. J., Lambeth, S. P., Mueller, J. L., & Townsend, S. W. (2020). Nonadjacent dependency processing in monkeys, apes, and humans. *Science Advances, 6,* eabb0725.

How about learning of **meaningful** sequences?



“ball fetch”
“stick point”

Ramos, D., & Ades, C. (2012). Two-item sentence comprehension by a dog (Canis familiaris). *PLoS ONE, 7,* e29689.



“to sugar take decoy”
“to decoy take sugar”

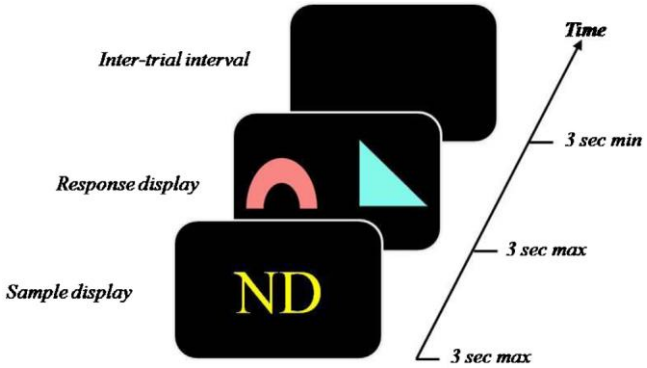
Pilley, J. W. (2013). Border collie comprehends sentences containing a prepositional object, verb, and direct object. *Learning and Motivation, 44,* 229-240.

[Chaser in action: https://www.youtube.com/watch?v=omaHv5sxiFI]

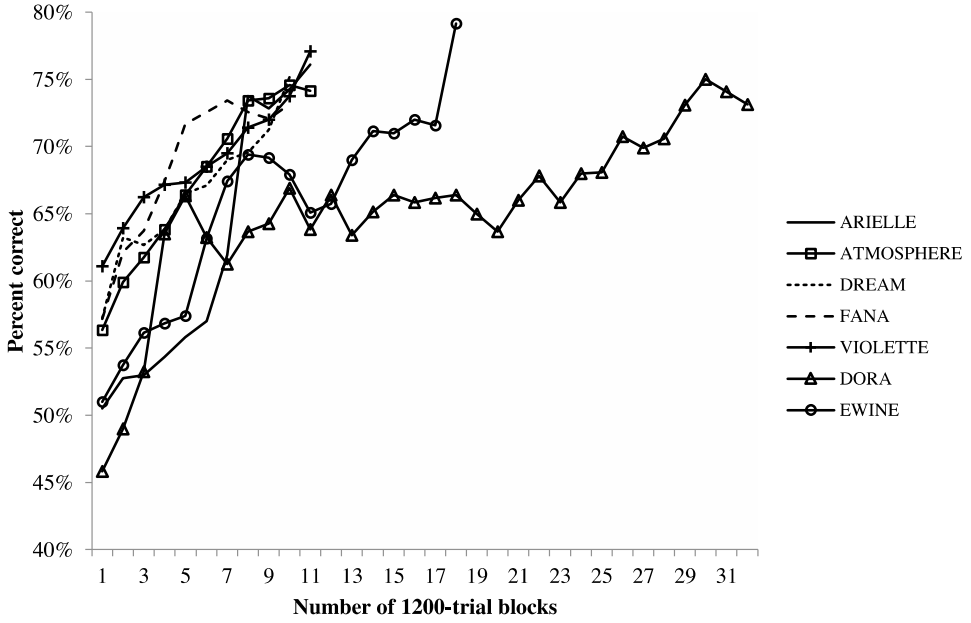
https://youtu.be/2Dhc2zePJFE

Savage-Rumbaugh, E. S., Murphy, J., Sevcik, R., Brakke, K., Williams, S., Rumbaugh, D., & Bates, E. (1993). Language comprehension in ape and child. *Monographs of the Society for Research in Child Development*, *58*, 1–252.

Puzzling failures in (most) baboons



6 letters (3 for shapes, 3 for colours)
3 shapes, 3 colours



Medam, T., & Fagot, J. (2016). Behavioral assessment of combinatorial semantics in baboons (Papio papio). *Behavior Processes, 123*, 54-62.

How about **production** of meaningful sequences?

Chase bite
Chase Kanzi
Person(g) come(g)
Tickle ball
Bite person(g)
Come(g) chase(g)
Ball tickle
Chase Sue
Kanzi chase
Surprise money
Bite chase
Pat(g) person(g)
Kanzi grab
Grab person(g)
Chase bite
Pat(g) this(g)
Chase come(g)
Person(g) go(g)
Ball pat(g)
Person(g) bite
Chase tickle

Chase person1(g) per
Person1(g) pat(g) per
Person1(g) person2(g

le 3

nparison of Most Frequent Comb

Kanzi
2 items
Chase person(g)
Person(g) chase(g)
Chase(g) Person(g)
Person(g) pat(g)
Chase bite
Chase Kanzi
Person(g) come(g)
Tickle ball
Bite person(g)
Come(g) chase(g)



Savage-Rumbaugh, E. S.,. McDonald, K., Sevcik, R. A., Hopkins, W. D., & Rupert, E. (1986). Spontaneous Symbol Acquisition and Communicative Use By Pygmy Chimpanzees (Pan paniscus). *Journal of Experimental Psychology: General, 115*, 211–235.



https://www.youtube.com/watch?v=UfHWTFlqEUE


FluentPet



https://www.youtube.com/watch?v=UfHWTFlqEUE

Summary on grammar learning

Artificial Grammar Learning suggests abilities to learn sequence constraints are present in other animals (including other primates)

- Grammars tested to date are quite simple
- Interpretation can be contentious

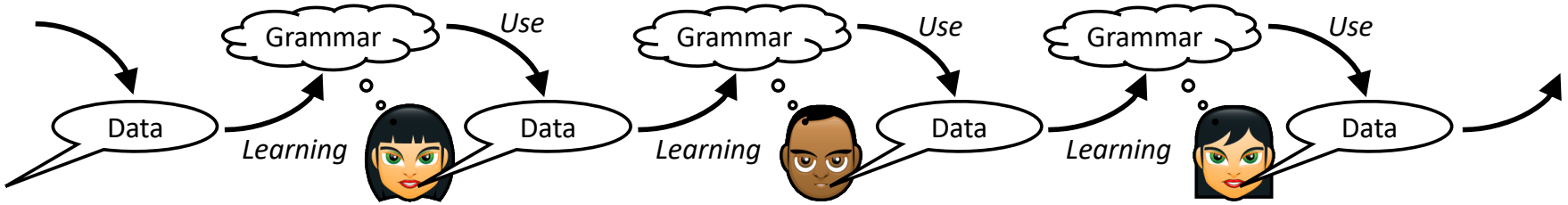
Language-trained animals can interpret complex (i.e. multi-part) expressions

- But larger-N lab studies surprisingly scarce, and these tasks seem to be hard

Humans are not unique in our ability to process meaningful sequences

- **But we may be uniquely proficient**

Reminder: Learning, use, and language design



- Language is passed from person to person by **learning**
- People learn from language as it is **used in communication**
- Language **evolves** in response to its learning and use

Reminder: the working hypothesis

- Humans ended up with an unusual combination of traits: ubiquitous social learning (including capacity for vocal learning and grammar learning) and **deep mental interpenetration**
- This set in place a cultural evolutionary process that shaped how language works

The Evolution of Social Cognition

Social cognition and language

Humans are unusual

- in our drive to share our mental states
- in our aptitude for reasoning about mental states in others

Social cognition and language

Humans are unusual

- **in our drive to share our mental states**
- in our aptitude for reasoning about mental states in others


Mitteilungsbedürfnis: A need
to share thoughts or feelings

Mitteilungsbedürfnis in non-human animals: the only recorded example

Wild chimpanzee 'holdout'/'show' gesture

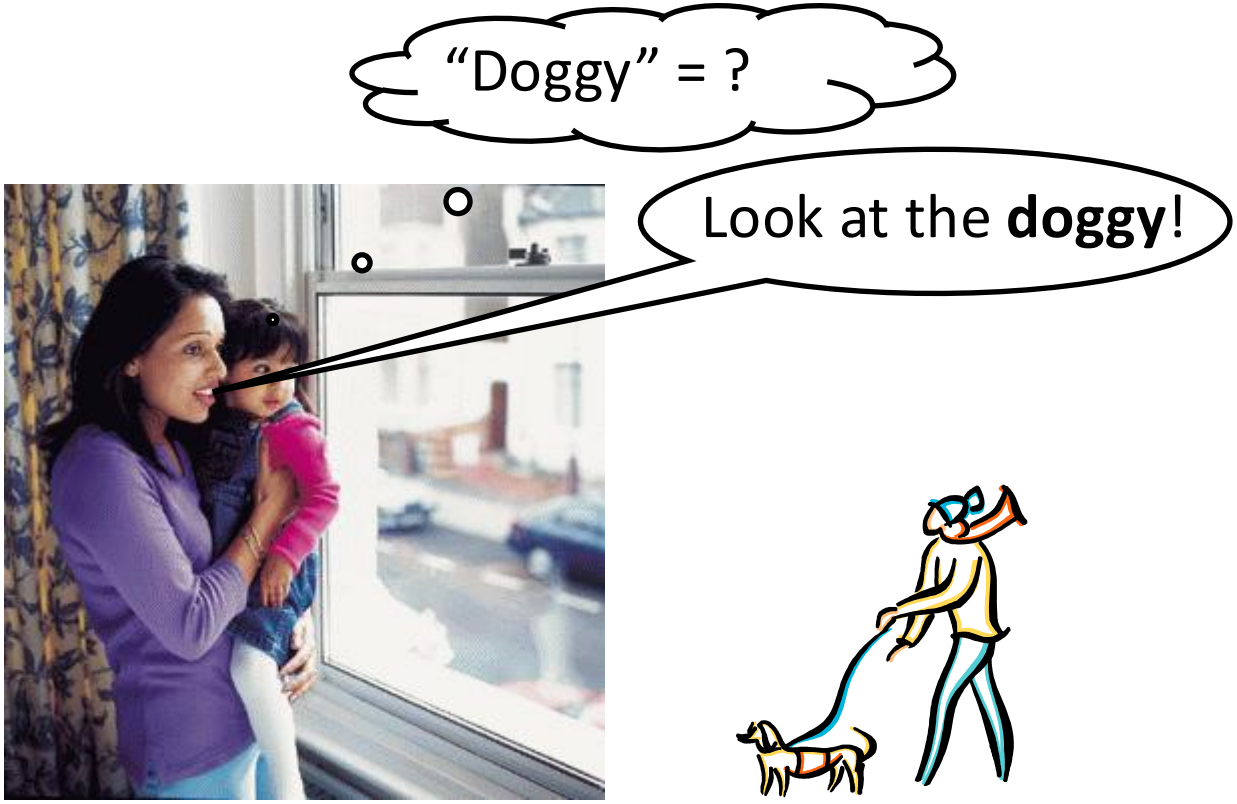
Wilke, C., Lahiff, N.J., Sabbi, K.H., Watts, D.P., Townsend, S.W., & Slocombe, K.E. (2022). Declarative referential gesturing in a wild chimpanzee (Pan troglodytes), *PNAS, 119*, e2206486119. https://doi.org/10.1073/pnas.2206486119.

Social cognition and language

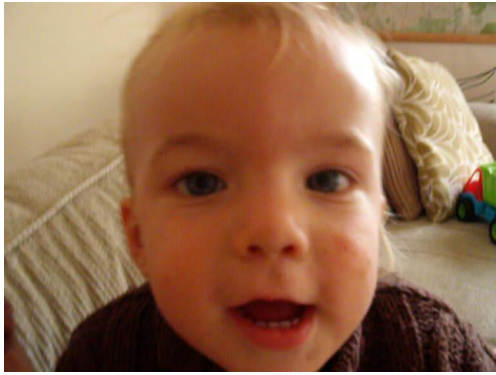
Humans are unusual

- in our drive to share our mental states
- **in our aptitude for reasoning about mental states in others**

Learning language involves inferring mental states of others

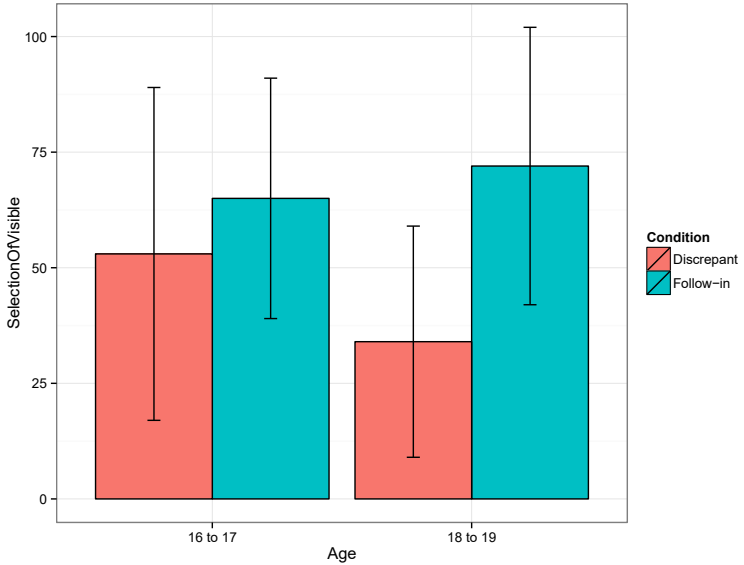


Exploiting attentional focus



Word learning, 16-19 month olds

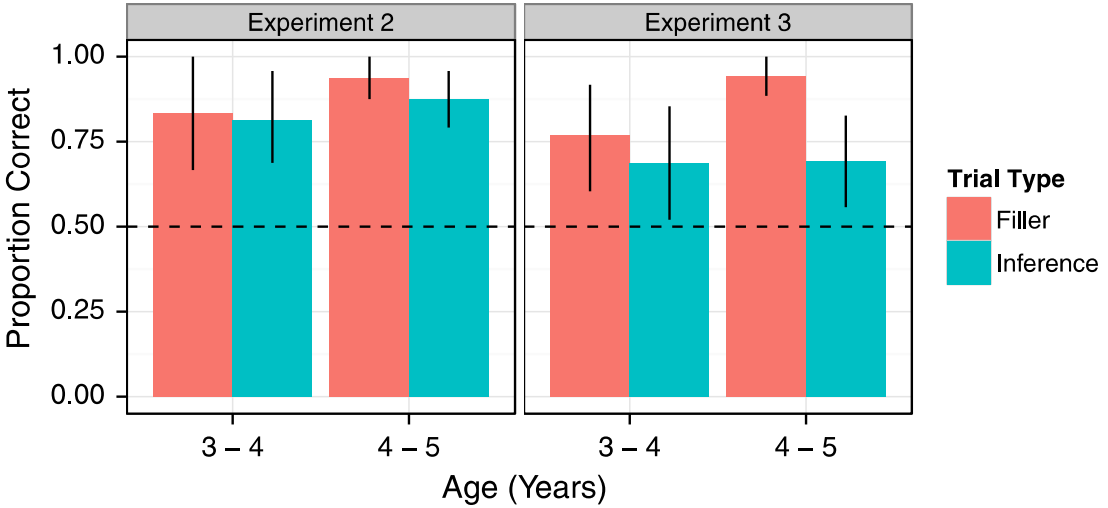
- Kid, experimenter, bucket, two novel objects
- Kid sees both toys, plays with one, other one goes back in the bucket
- Follow-in labelling: experimenter looks at toy kid is looking at and labels it (“it’s a toma!”)
- **Discrepant** labelling: experimenter looks at toy in bucket and labels it (“it’s a toma!”)

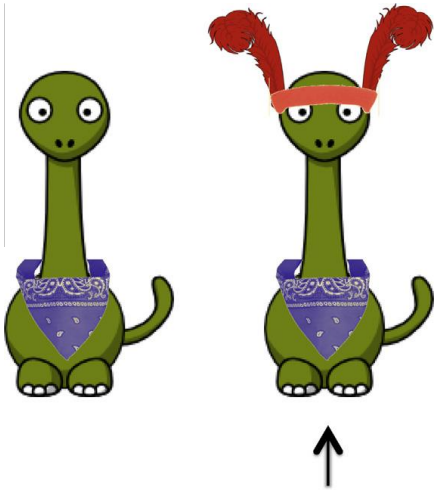


Baldwin, D. A. (1991). Infants’ contribution to the achievement of joint reference. *Child Development, 62,* 875–890.

Expectations about how people use words

Do children assume that people use words in an informative way?





"This is a dinosaur with a dax" (Exp 2)
"Here is a dinosaur with a dax" (Exp 3)

Frank, M. C., & Goodman, N. D. (2014). Inferring word meanings by assuming that speakers are informative. *Cognitive Psychology, 75,* 80-96.

Using language involves inferring mental states of others (ostensive-inferential communication)

Speaker's utterances (or other communicative behaviours)

- provide evidence about their thoughts (**ostension**)
- are designed to allow the hearer to infer those thoughts (**inference**)

Hearers infer meaning, guided by the knowledge that the speaker wants the hearer to be able to infer their meaning

Using language involves inferring mental states of others (ostensive-inferential communication)

The Cooperative Principle and Gricean Maxims

- **Quality:** Be truthful
- **Quantity:** Be as informative as required
- **Relation:** Be relevant
- **Manner:** Be clear

A: Where's Bill?

B: His dog died

The usual question: how did capacity to reason about mental states in others evolve?

Is it a human-unique trait?

Or can we see similar abilities in our closest living relatives?

Reminder: Absence of 2nd order intentional communication in macaques?

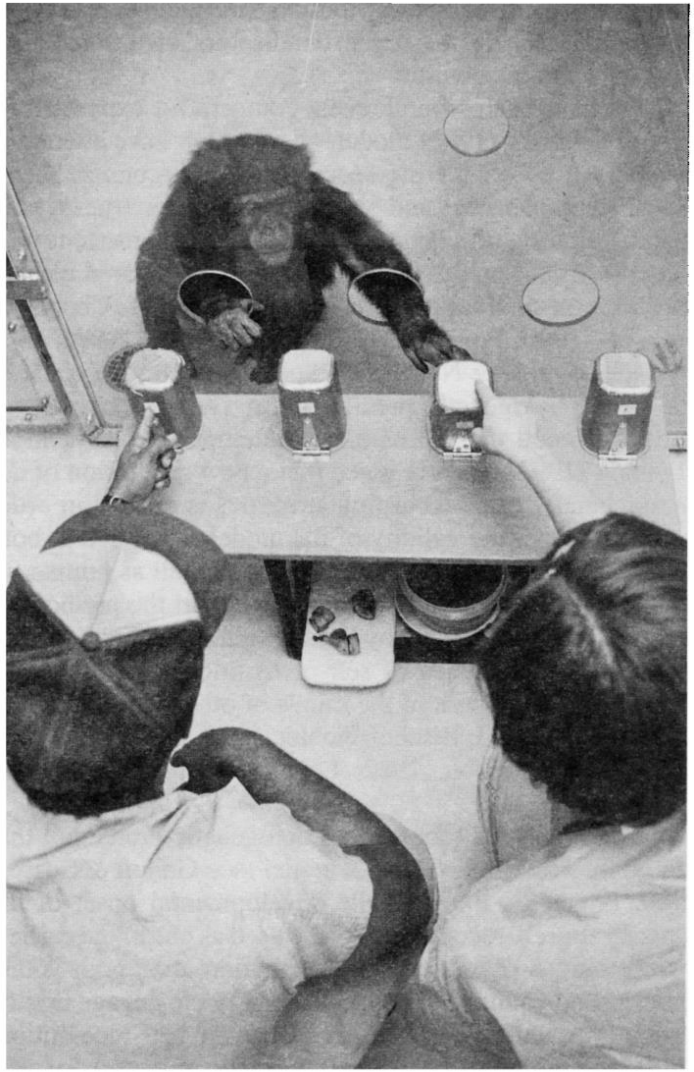
- Mothers and infants
- **Ignorance condition**: Mother knows something, infant doesn't
  - Presence of food, predator
- **Knowledge condition:** They both know it
- **Mothers' vocalizations didn't differ between conditions**





Cheney, D., & Seyfarth, R. (1990). Attending to behaviour versus attending to knowledge: examining monkeys' attribution of mental states. *Animal Behavior, 40,* 742-753.


WILD HD
Other apes seem to struggle to understand
how cooperative communication works



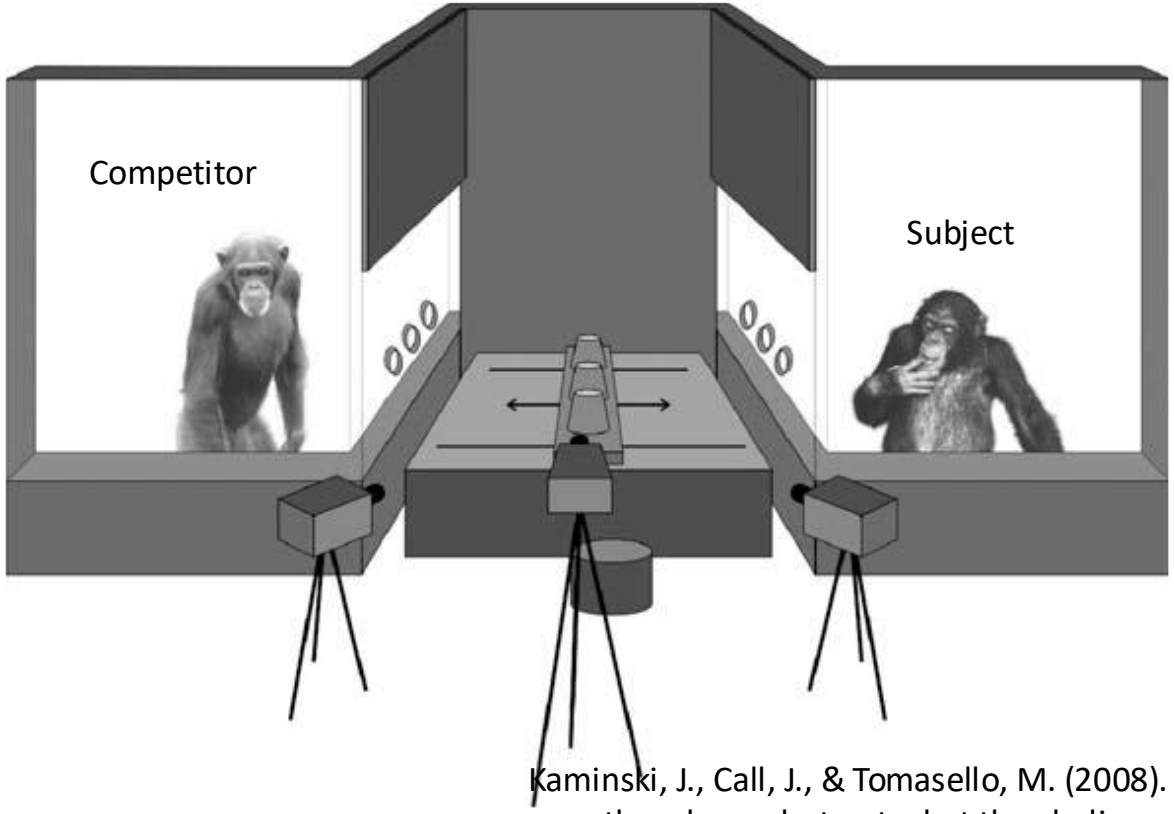
Knowing others' minds: knowing what others know

- 6 juvenile chimps (approx. 4 y. o.)
- Two experimenters
- "Guesser" leaves room
- "Knower" hides food under cup
  - Chimp can't see which one
- Both humans point to a cup
- Chimp indicates which cup he wants to look under

Kids can do this age 4, **chimps at chance**

Povinelli, D. J., Rulf, A. B., & Bierschwale, D. T. (1994). Absence of knowledge attribution and self-recognition in young chimpanzees (*Pan troglodytes*). *Journal of Comparative Psychology, 108,* 74–80.

Knowing others' minds: knowing what others know



Kaminski, J., Call, J., & Tomasello, M. (2008). Chimpanzees know what others know, but not what they believe. *Cognition, 109*, 224–234.

wooclap.com, code INMRBS

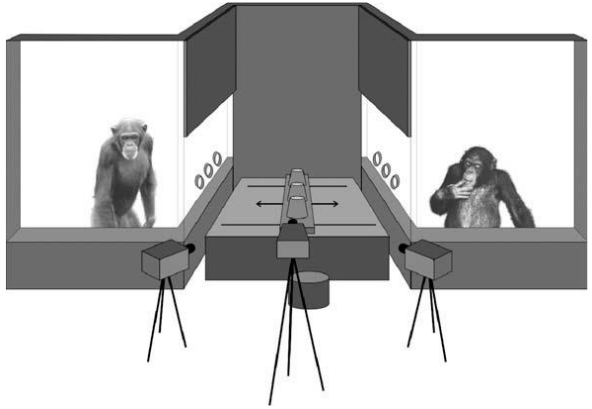
Reward A that you both saw hidden

Reward B that **only you** saw hidden

Competitor selects first, you don't see what they did

When it's your turn, which do you go for?

1. Reward A
2. Reward B



wooclap.com, code INMRBS

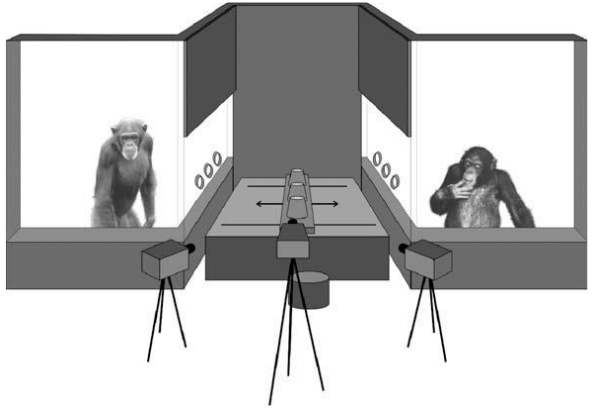
Reward A that you both saw hidden

Reward B that only you saw hidden

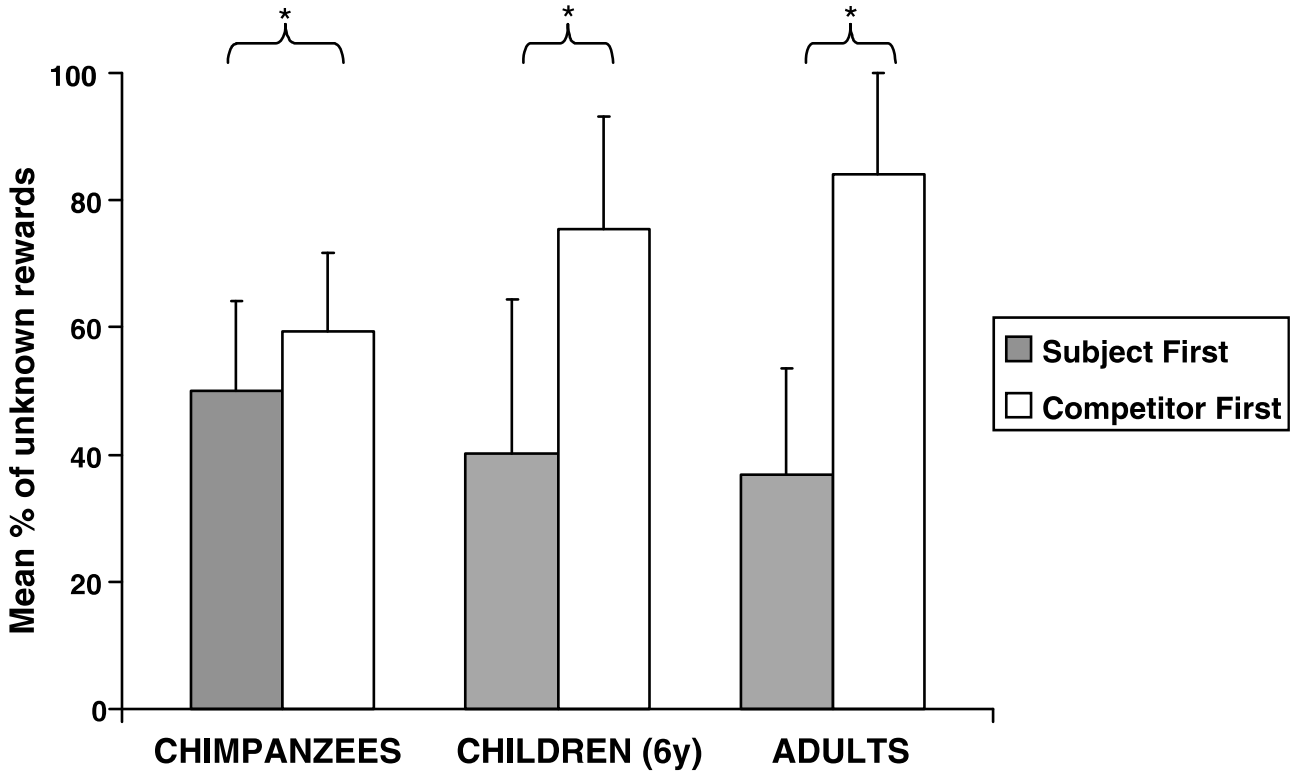
You select first, competitor won't see what you did

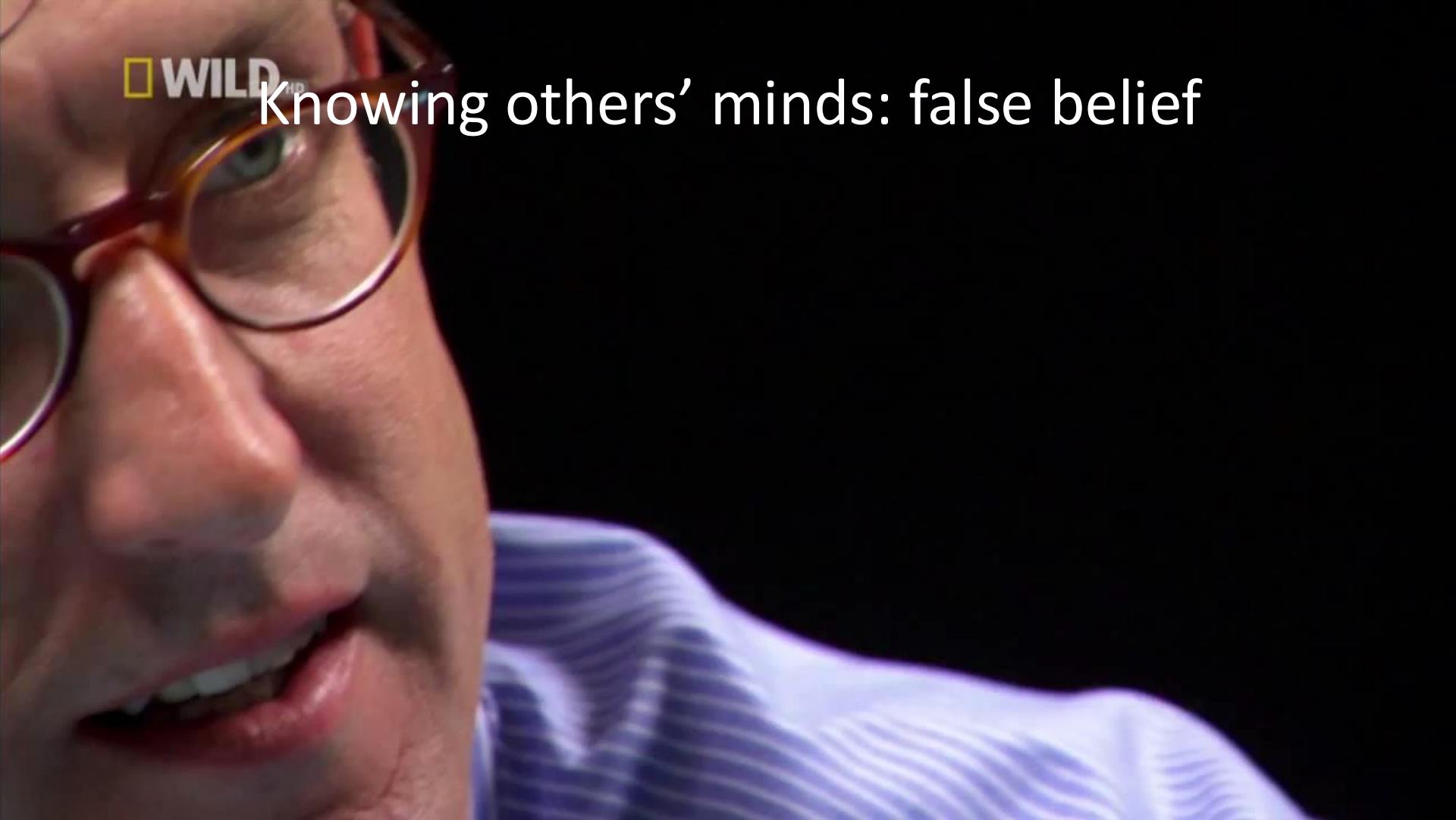
Which do you go for?

1. Reward A
2. Reward B



Knowing others' minds: knowing what others know





Knowing others' minds: false belief

A question from the pre-lecture quiz

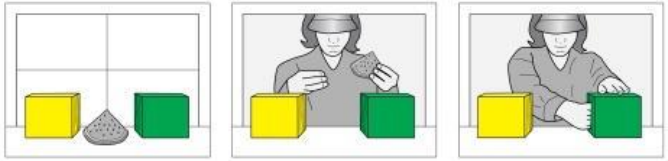
`I had trouble answering question 4 [Why is the ability to understand false belief relevant to second-order intentionality in communication?], and I think the problem already lies in my understanding of the definition of "false belief". I think there's a difference between these two explanations from the paper: 1) False belief "requires an understanding that other minds contain representations of the world *that can be different from reality*". and 2) False belief requires individuals to "represent another's mental state in a way that is *independent from their own representation of reality*". I actually prefer the second explanation because phrased like this, it is consistent with the idea that we might not be able to perceive an objective reality (if there is one at all), because we will only ever be able to see our representation of reality. Rather than defining false belief as "holding the belief that someone else has a belief that *you know is not true*", I would then opt for something along the lines of "holding the belief that someone else has a belief that *does not align with yours*".'

A question from the pre-lecture quiz

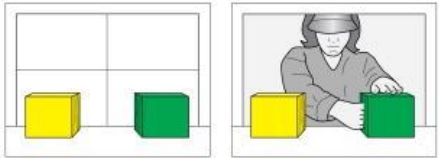
`Apart from my confusion about the definition of false belief, I'm also not quite sure how false belief relates to second-order intentionality in communication. Why would it be necessary for an individual to believe someone else's belief to be false in order to reason about their communicative intentions? Wouldn't it just be enough for an individual to recognise that someone else has beliefs/representations/intentions that may differ from their own (just because they're different, they don't have to be false, or judged to be false)? Or is it just that if an individual believes someone else's beliefs to be false, we basically have definite proof that the individual REALLY understands that someone else's beliefs are different from their own?’

Familiarization trials

Trial 1



Trials 2 and 3



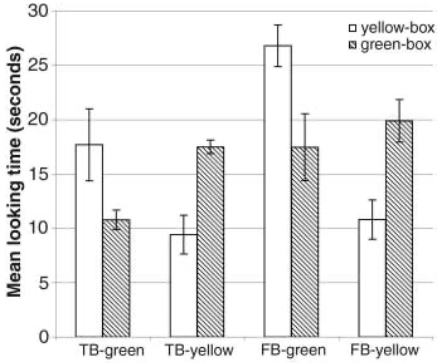
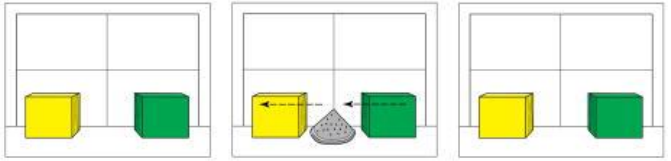


Fig. 4. Mean (±SE) looking times during the test trial (after the actor reached into the green or yellow box) in the four belief conditions.

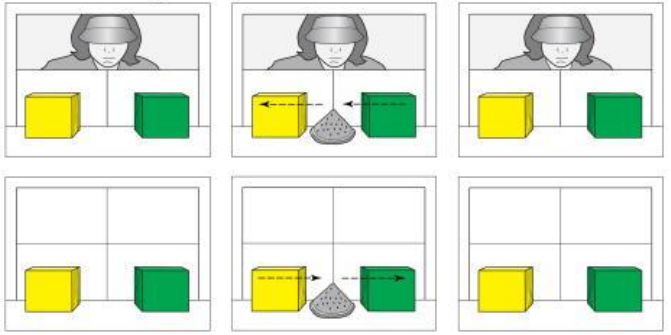
Onishi, K. H., & Baillargeon, R. (2005). Do 15-month-old infants understand false beliefs? *Science, 308*, 255–258

Belief-induction trial

False-belief-green condition

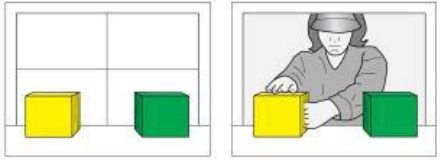


False-belief-yellow condition

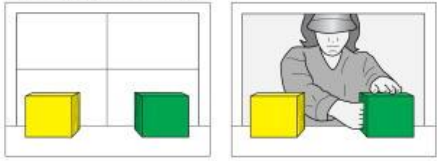


Test trial

Yellow-box event



Green-box event



False-belief 1

Chimpanzee Hatsuka

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

Familiarization

Bonobo Jasongo

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

Knowing others' minds: false belief

Table 1. Number of participants who made first looks to either the target or the distractor during the agent's approach in experiments one (N = 40) and two (N = 30). Values in parentheses indicate the number of participants who did not look at either.

Condition	Target	Distractor	Total
	Experiment one		
FB1	10	4	14 (6)
FB2	10	6	16 (4)
Total	20	10	30 (10)
	Experiment two		
FB1	8	2	10 (6)
FB2	9	3	12 (2)
Total	17	5	22 (8)

Krupenye, C., et al. (2016). Great apes anticipate that other individuals will act according to false beliefs. *Science, 354,* 110-114.

So why are we so good at it?

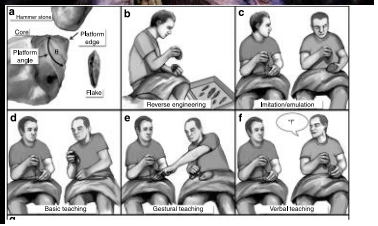
What selective pressures drive the evolution of mind sharing and mind reading?

- We occupy a uniquely social niche?
- We occupy a uniquely technological niche?
- …









Reminder: communication in the Last Common Ancestor of chimps and humans

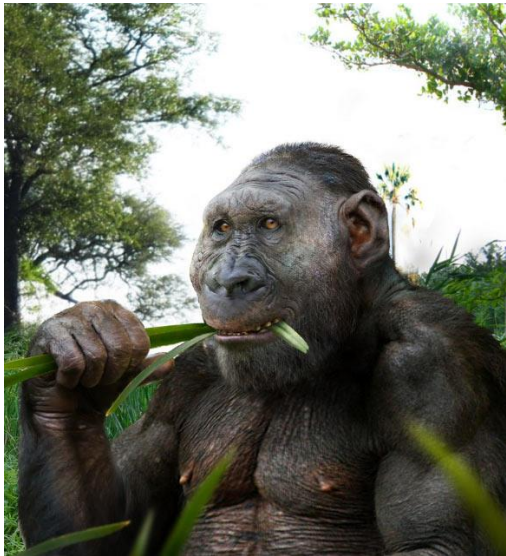
- Not 2nd order intentional?
- No/minimal use of structure subserving meaning?
- Probably not learned?

But remember Fitch's point: their communication system may underrepresent their cognitive capacities!



How about these other capacities?

- The ability to learn complex grammars?
  - No/little capacity for complex vocal imitation
  - Some ability to learn complex sequencing constraints?
  - Some ability to learn compositional meaning-form mappings?
- Little motivation to mindshare??
- Some ability to mindread



The human package

Somehow, we ended up with

- The ability to learn complex grammars
  - capacity for complex vocal imitation
  - ability to learn complex sequencing constraints
  - ability to learn compositional meaning-form mappings
- The ability and motivation to mindshare and mindread

This sets up the preconditions for the **cultural transmission of learned, meaning-bearing communication**

- Once that's in place, exciting stuff happens

Next up

- Workshop
  - Animal grammar learning, a linguist takes a look
- Next lecture: the cultural evolution of grammar